12TH CHEMISTRY · CHAPTER 29 · FULL CHAPTER MCQS

Spectroscopy–1

119 topic-wise MCQs from the complete chapter with answers and reasons

ELECTROMAGNETIC RADIATION

1. Electromagnetic waves consist of oscillating:

(a) electric fields only
(b) electric and magnetic fields
(c) magnetic fields only
(d) sound waves

Answer: (b) electric and magnetic fields — Stated in the introduction.

2. The speed of light is:

(a) 3.0×10^{10} m s^{-1}
(b) 3.0×10^{6} m s^{-1}
(c) 3.0×10^{8} m s^{-1}
(d) 6.626×10^{-34} m s^{-1}

Answer: (c) 3.0×10^{8} m s^{-1} — c = 3.0×10^{8} m/s.

3. The relation between speed, wavelength and frequency is:

(a) $c = \lambda\nu$
(b) $c = \lambda/\nu$
(c) $c = \nu/\lambda$
(d) $c = \lambda + \nu$

Answer: (a) $c = \lambda\nu$ — Given in the book.

4. The energy of radiation is given by:

(a) $E = \nu/h$
(b) $E = h/\nu$
(c) $E = h\lambda$
(d) $E = h\nu$

Answer: (d) $E = h\nu$ — Planck's equation.

5. Planck's constant is:

(a) 6.626×10^{-34} J s
(b) 6.022×10^{23} J s
(c) 3.0×10^{8} J s
(d) 1.6×10^{-19} J s

Answer: (a) 6.626×10^{-34} J s — Given in the book.

6. Energy of radiation is directly proportional to its:

(a) wavelength
(b) frequency
(c) amplitude
(d) speed

Answer: (b) frequency — E = hν.

7. The radiation with the highest energy is:

(a) infrared
(b) radio waves
(c) microwaves
(d) gamma rays

Answer: (d) gamma rays — Shortest wavelength.

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8. The radiation with the lowest energy is:

(a) ultraviolet (b) X-rays
(c) radio waves (d) visible light

Answer: (c) radio waves — Longest wavelength.

9. The visible region is about:

(a) 1–10 nm (b) 100–200 nm
(c) 800–1500 nm (d) 400–750 nm

Answer: (d) 400–750 nm — From Figure 29.1.

10. The unit of frequency is:

(a) metre (b) Hertz (s^{-1})
(c) joule (d) nm

Answer: (b) Hertz (s^{-1}) — Given in the book.

11. The smallest quantity of radiation that can be emitted or absorbed is a:

(a) quantum (b) mole
(c) wave (d) spectrum

Answer: (a) quantum — Planck's quantum theory.

29.1 INDEX OF HYDROGEN DEFICIENCY

12. IHD measures the:

(a) molecular mass (b) degrees of unsaturation
(c) boiling point (d) number of isotopes

Answer: (b) degrees of unsaturation — Definition.

13. One degree of unsaturation corresponds to a loss of:

(a) two hydrogen atoms (b) one hydrogen atom
(c) four hydrogen atoms (d) one carbon atom

Answer: (a) two hydrogen atoms — Per pair of H missing.

14. The IHD of an open-chain alkane is:

(a) 2 (b) 1
(c) 0 (d) 3

Answer: (c) 0 — Formula C_nH_{2n+2}.

15. A compound with IHD = 1 may have:

(a) no ring and no double bond (b) one triple bond
(c) two rings (d) one double bond or one ring

Answer: (d) one double bond or one ring — Table 29.1.

16. A triple bond counts as an IHD of:

(a) 3 (b) 1
(c) 2 (d) 0

Answer: (c) 2 — Keep in Mind box.

17. A ring counts as an IHD of:

(a) 1 (b) 0
(c) 2 (d) 3

Answer: (a) 1 — Same as a double bond.

18. The IHD of 1-hexene, C_6H_{12}, is:

(a) 3 (b) 0
(c) 2 (d) 1

Answer: (d) 1 — (14 − 12) ÷ 2.

19. The IHD of cyclohexane, C_6H_{12}, is:

(a) 0 (b) 1
(c) 2 (d) 6

Answer: (b) 1 — One ring.

20. The IHD of benzene, C_6H_6, is:

(a) 2 (b) 3
(c) 6 (d) 4

Answer: (d) 4 — (14 − 6) ÷ 2; one ring + three double bonds.

21. The IHD of C_4H_6 is:

(a) 2 (b) 1
(c) 3 (d) 4

Answer: (a) 2 — (10 − 6) ÷ 2.

22. The IHD of C_4H_4 is:

(a) 4 (b) 2
(c) 3 (d) 1

Answer: (c) 3 — (10 − 4) ÷ 2.

23. A compound with IHD = 2 can NOT be a compound with:

(a) two rings (b) three double bonds
(c) one triple bond (d) one ring and one double bond

Answer: (b) three double bonds — Three double bonds need IHD 3.

24. 1-Hexene and cyclohexane are:

(a) allotropes (b) isotopes
(c) polymers (d) isomers

Answer: (d) isomers — Same molecular formula, different structures.

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25. A hydrocarbon with IHD = 0 is:

(a) an alkyne (b) aromatic
(c) saturated (d) cyclic

Answer: (c) saturated — No rings or multiple bonds.

26. IHD can be found from the:

(a) molecular formula alone (b) IR spectrum only
(c) mass spectrum only (d) melting point

Answer: (a) molecular formula alone — No complex technique is needed.

29.2 ATOMIC SPECTRUM

27. Atomic spectra are evidence that electrons occupy:

(a) the nucleus (b) continuous energy bands
(c) discrete energy levels (d) random positions

Answer: (c) discrete energy levels — Stated in the book.

28. An emission spectrum is produced by:

(a) dissolving a salt
(b) cooling a liquid
(c) passing white light through a cool gas
(d) heating a low-pressure gas

Answer: (d) heating a low-pressure gas — Electrons are excited and fall back.

29. An emission spectrum appears as:

(a) dark lines on a coloured background
(b) coloured lines on a black background
(c) a continuous rainbow
(d) a single white band

Answer: (b) coloured lines on a black background — Stated in the book.

30. An absorption spectrum appears as:

(a) dark lines on a coloured background
(b) coloured lines on a black background
(c) a black band only
(d) a white band only

Answer: (a) dark lines on a coloured background — Absorbed wavelengths are missing.

31. An absorption spectrum is produced by passing white light through:

(a) a vacuum (b) a hot solid
(c) a prism only (d) a cool, low-pressure gas

Answer: (d) a cool, low-pressure gas — Stated in the book.

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32. When an excited electron returns to a lower level it:

(a) emits a photon (b) absorbs a photon
(c) leaves the atom (d) enters the nucleus

Answer: (a) emits a photon — This gives a spectral line.

33. An absorption spectrum is like a _____ of the emission spectrum.

(a) magnified copy (b) mirror image
(c) photographic negative (d) continuous version

Answer: (c) photographic negative — Lines are at the same positions.

34. Each spectral line corresponds to a specific:

(a) mass (b) wavelength
(c) volume (d) pressure

Answer: (b) wavelength — One transition, one wavelength.

35. A line of the hydrogen emission spectrum in the visible region is at:

(a) 217 nm (b) 589 nm
(c) 656 nm (d) 100 nm

Answer: (c) 656 nm — The red line.

29.3 SPECTROSCOPY

36. Spectroscopy studies the interaction of matter with:

(a) magnetism only (b) sound
(c) gravity (d) electromagnetic radiation

Answer: (d) electromagnetic radiation — Definition.

37. Identifying an unknown sample is called:

(a) quantitative analysis (b) qualitative analysis
(c) gravimetric analysis (d) volumetric analysis

Answer: (b) qualitative analysis — Stated in the book.

38. Measuring the amount of a substance is called:

(a) quantitative analysis (b) qualitative analysis
(c) fragmentation (d) conjugation

Answer: (a) quantitative analysis — Stated in the book.

39. Emission line spectra are unique to each element, like a:

(a) mirror (b) shadow
(c) fingerprint (d) magnet

Answer: (c) fingerprint — No two elements give the same lines.

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40. The yellow flame of sodium is due to lines at:

(a) 217 nm (b) 656 nm
(c) 450 nm (d) 589 nm

Answer: (d) 589 nm — Stated in the book.

41. Most emission lines of mercury are below 450 nm, giving:

(a) blue light (b) red light
(c) yellow light (d) green light

Answer: (a) blue light — Stated in the book.

42. Astronomers find the composition of stars by using:

(a) chromatography (b) atomic spectra
(c) titration (d) IHD

Answer: (b) atomic spectra — Did You Know box.

29.4 UV/VISIBLE SPECTROSCOPY

43. Absorbance and transmittance are related by:

(a) A = 1/T (b) A = log T
(c) A = T (d) A = −log T

Answer: (d) A = −log T — Given in the book.

44. UV-Vis spectroscopy studies changes in:

(a) nuclear spin (b) bond vibrations
(c) electronic energy levels (d) isotopic mass

Answer: (c) electronic energy levels — Electrons are excited.

45. π electrons are present only in:

(a) alkanes (b) unsaturated compounds
(c) noble gases (d) all compounds

Answer: (b) unsaturated compounds — Stated in the book.

46. The lowest-energy orbital among these is:

(a) σ bonding (b) π bonding
(c) non-bonding (d) π*

Answer: (a) σ bonding — σ < π < n.

47. The highest-energy filled orbital is usually:

(a) π bonding (b) σ bonding
(c) non-bonding (n) (d) σ*

Answer: (c) non-bonding (n) — Lone pairs.

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48. The orbitals that are normally empty are:

(a) σ and π (b) σ* and π*
(c) n and π (d) σ and n

Answer: (b) σ* and π* — Anti-bonding orbitals.

49. The transition requiring the most energy is:

(a) σ → σ* (b) n → π*
(c) π → π* (d) n → σ*

Answer: (a) σ → σ* — Largest gap.

50. The transition requiring the least energy is:

(a) n → σ* (b) σ → σ*
(c) π → π* (d) n → π*

Answer: (d) n → π* — Smallest gap.

51. The wavelength of maximum absorption is called:

(a) m/e (b) λ_{max}
(c) IHD (d) R_f

Answer: (b) λ_{max} — Definition.

52. λ_{max} of buta-1,3-diene is:

(a) 290 nm (b) 257 nm
(c) 217 nm (d) 656 nm

Answer: (c) 217 nm — Figure 29.5.

53. Buta-1,3-diene is colourless because it absorbs in the:

(a) radio region (b) visible region
(c) infrared region (d) UV region

Answer: (d) UV region — 217 nm is not visible.

54. The only transition in buta-1,3-diene (in range) is:

(a) π → π* (b) n → π*
(c) n → σ* (d) σ → n

Answer: (a) π → π* — It has no non-bonding electrons.

55. λ_{max} of hexa-1,3,5-triene is about:

(a) 190 nm (b) 217 nm
(c) 257 nm (d) 589 nm

Answer: (c) 257 nm — More conjugation.

56. As conjugation increases, λ_{max}:

(a) increases (b) decreases
(c) stays constant (d) becomes zero

Answer: (a) increases — Energy gaps become smaller.

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57. Each extra conjugated double bond raises λ_{max} by about:

(a) 1–2 nm (b) 30–40 nm
(c) 200 nm (d) 500 nm

Answer: (b) 30–40 nm — Stated in the book.

58. Highly conjugated compounds are often:

(a) IR inactive (b) colourless
(c) gases (d) brightly coloured

Answer: (d) brightly coloured — They absorb visible light.

59. Longer wavelength light has:

(a) more energy (b) less energy
(c) higher frequency (d) the same energy

Answer: (b) less energy — E = hc/λ.

60. The red colour of ripe tomato is due to:

(a) lycopene (b) chlorophyll
(c) haemoglobin (d) carotene only

Answer: (a) lycopene — Interesting Information box.

61. Green tomato contains more:

(a) iodine (b) lycopene
(c) chlorophyll (d) ninhydrin

Answer: (c) chlorophyll — Interesting Information box.

62. Which compound shows no significant absorption at 200–800 nm?

(a) but-3-en-2-one (b) buta-1,3-diene
(c) hexa-1,3,5-triene (d) propan-2-ol

Answer: (d) propan-2-ol — It has no π electrons.

63. Ethene can show which transition?

(a) π → π* only (b) n → π* only
(c) n → σ* only (d) none

Answer: (a) π → π* only — No lone pairs.

29.5 INFRARED SPECTROSCOPY

64. IR spectroscopy is based on changes in:

(a) isotopes (b) electronic levels
(c) nuclear mass (d) vibrations of bonds

Answer: (d) vibrations of bonds — Bonds act like springs.

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65. A molecule absorbs IR only if it has:

(a) no dipole
(b) a dipole that changes as it vibrates
(c) an unpaired electron
(d) a ring

Answer: (b) a dipole that changes as it vibrates — Stated in the book.

66. Which molecule is IR inactive?

(a) HCl
(b) H_2O
(c) O_2
(d) CO

Answer: (c) O_2 — Symmetrical and non-polar.

67. Which is NOT a mode of vibration shown in the book?

(a) rotational spinning
(b) symmetric stretching
(c) asymmetric stretching
(d) symmetric bending

Answer: (a) rotational spinning — Figure 29.6 shows four modes.

68. The fingerprint region lies below:

(a) 3000 cm^{-1}
(b) 4000 cm^{-1}
(c) 1500 cm^{-1}
(d) 2500 cm^{-1}

Answer: (c) 1500 cm^{-1} — Stated in the book.

69. The functional group region lies above:

(a) 1500 cm^{-1}
(b) 500 cm^{-1}
(c) 100 cm^{-1}
(d) 1000 cm^{-1}

Answer: (a) 1500 cm^{-1} — Stated in the book.

70. Stronger bonds absorb at:

(a) the same wavenumber
(b) lower wavenumbers
(c) zero wavenumber
(d) higher wavenumbers

Answer: (d) higher wavenumbers — They need higher energy.

71. The C–O bond absorbs at:

(a) 1500–1680 cm^{-1}
(b) 1040–1300 cm^{-1}
(c) 2200–2250 cm^{-1}
(d) 3300–3500 cm^{-1}

Answer: (b) 1040–1300 cm^{-1} — Table 29.2.

72. The C=C bond absorbs at:

(a) 1500–1680 cm^{-1}
(b) 1040–1300 cm^{-1}
(c) 2850–2950 cm^{-1}
(d) 3200–3600 cm^{-1}

Answer: (a) 1500–1680 cm^{-1} — Table 29.2.

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73. The C≡N bond of a nitrile absorbs at:

(a) 2850–2950 cm^{-1}
(b) 1640–1690 cm^{-1}
(c) 2200–2250 cm^{-1}
(d) 3300–3500 cm^{-1}

Answer: (c) 2200–2250 cm^{-1} — Table 29.2.

74. The C–H bond of an alkane absorbs at:

(a) 2200–2250 cm^{-1}
(b) 1040–1300 cm^{-1}
(c) 1670–1740 cm^{-1}
(d) 2850–2950 cm^{-1}

Answer: (d) 2850–2950 cm^{-1} — Table 29.2.

75. The N–H bond absorbs at:

(a) 1500–1680 cm^{-1}
(b) 3300–3500 cm^{-1}
(c) 2500–3300 cm^{-1}
(d) 1040–1300 cm^{-1}

Answer: (b) 3300–3500 cm^{-1} — Table 29.2.

76. The O–H bond of an alcohol absorbs at:

(a) 3200–3600 cm^{-1}
(b) 1710–1750 cm^{-1}
(c) 2200–2250 cm^{-1}
(d) 1040–1300 cm^{-1}

Answer: (a) 3200–3600 cm^{-1} — Table 29.2.

77. The O–H bond of a carboxylic acid absorbs at:

(a) 1500–1680 cm^{-1}
(b) 3600–4000 cm^{-1}
(c) 2500–3300 cm^{-1}
(d) 1040–1300 cm^{-1}

Answer: (c) 2500–3300 cm^{-1} — A very broad band.

78. The C=O bond of an ester absorbs at:

(a) 3200–3600 cm^{-1}
(b) 1640–1690 cm^{-1}
(c) 2200–2250 cm^{-1}
(d) 1710–1750 cm^{-1}

Answer: (d) 1710–1750 cm^{-1} — Table 29.2.

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79. The C=O bond of an amide absorbs at:

(a) 1710–1750 cm^{-1}
(b) 1640–1690 cm^{-1}
(c) 2850–2950 cm^{-1}
(d) 3300–3500 cm^{-1}

Answer: (b) 1640–1690 cm^{-1} — Table 29.2.

80. A strong sharp peak near 1710 cm^{-1} indicates:

(a) an N–H group (b) an O–H group
(c) a C≡N group (d) a C=O group

Answer: (d) a C=O group — Example 29.1.

81. A primary amine shows N–H absorption as:

(a) a peak at 2250 cm^{-1}
(b) one strong peak at 1700 cm^{-1}
(c) two weak peaks at 3300 and 3400 cm^{-1}
(d) no peak at all

Answer: (c) two weak peaks at 3300 and 3400 cm^{-1} — Stated in the book.

82. A broad band at 2500–3300 with a peak at 1700 cm^{-1} shows:

(a) a carboxylic acid (b) an alkane
(c) a nitrile (d) an amine

Answer: (a) a carboxylic acid — O–H and C=O together.

83. Compared with O–H of alcohols, the N–H signal of amines is:

(a) broader (b) less broad
(c) absent (d) at 1700 cm^{-1}

Answer: (b) less broad — Note in the book.

84. A roadside breathalyser detects ethanol using:

(a) infrared spectroscopy (b) mass spectrometry
(c) UV light (d) chromatography

Answer: (a) infrared spectroscopy — Did You Know box.

85. IR sensors on motorways measure:

(a) noise (b) speed of vehicles
(c) pollutants in vehicle emissions (d) humidity

Answer: (c) pollutants in vehicle emissions — CO, CO_2 and unburnt hydrocarbons.

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86. Propanone and propan-2-ol are distinguished in IR by:

(a) their M+2 peaks
(b) their colour
(c) their C–H peaks only
(d) C=O at 1710 vs O–H at 3200–3600 cm^{-1}

Answer: (d) C=O at 1710 vs O–H at 3200–3600 cm^{-1} — Example 29.1.

29.6 MASS SPECTROMETRY

87. In mass spectrometry the sample is bombarded with:

(a) neutrons (b) protons
(c) infrared light (d) high-energy electrons

Answer: (d) high-energy electrons — Stated in the book.

88. The ions detected in mass spectrometry are:

(a) positive ions (b) negative ions
(c) neutral atoms (d) free radicals only

Answer: (a) positive ions — An electron is lost.

89. m/e stands for:

(a) mass to energy ratio (b) mole to electron ratio
(c) mass to charge ratio (d) metal to element ratio

Answer: (c) mass to charge ratio — Definition.

90. The heights of peaks for isotopes show their:

(a) density (b) relative abundance
(c) radius (d) boiling points

Answer: (b) relative abundance — Stated in the book.

91. The relative abundances of Cl-35 and Cl-37 are:

(a) 75% and 25% (b) 50% and 50%
(c) 25% and 75% (d) 99% and 1%

Answer: (a) 75% and 25% — Stated in the book.

92. The relative abundance of boron-11 is:

(a) 19.9% (b) 80.1%
(c) 50% (d) 1.1%

Answer: (b) 80.1% — Mass spectrum of boron.

93. The relative atomic mass of boron from its mass spectrum is:

(a) 11.00 (b) 10.00
(c) 10.80 (d) 10.20

Answer: (c) 10.80 — (19.9 × 10 + 80.1 × 11) ÷ 100.

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94. The relative atomic mass of oxygen (to 2 d.p.) is:

(a) 17.00 (b) 15.90
(c) 16.50 (d) 16.00

Answer: (d) 16.00 — Example 29.2.

95. The molecular ion is the molecule that has:

(a) gained one electron (b) lost one electron
(c) lost one proton (d) broken into two

Answer: (b) lost one electron — Stated in the book.

96. The molecular ion peak gives the:

(a) number of rings (b) boiling point
(c) molecular mass (d) bond angle

Answer: (c) molecular mass — Usually the highest m/e.

97. The molecular ion has:

(a) a positive charge and one unpaired electron
(b) a negative charge
(c) no charge
(d) two unpaired neutrons

Answer: (a) a positive charge and one unpaired electron — One electron of a pair is removed.

98. The [M+1] peak is due to:

(a) oxygen-18 (b) chlorine-37
(c) bromine-81 (d) carbon-13

Answer: (d) carbon-13 — About 1.1% of carbon.

99. The natural abundance of carbon-13 is about:

(a) 11% (b) 1.1%
(c) 25% (d) 50%

Answer: (b) 1.1% — Stated in the book.

100. The [M+1] peak is larger for:

(a) none; it is constant (b) ethane than hexane
(c) methane than hexane (d) hexane than ethane

Answer: (d) hexane than ethane — More carbon atoms.

101. M abundance 85, [M+1] abundance 3: the number of carbon atoms is:

(a) 3 (b) 1
(c) 5 (d) 8

Answer: (a) 3 — (100 × 3) ÷ (1.1 × 85) = 3.21.

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102. The molecular ion of propanal appears at m/e:

(a) 44 (b) 72
(c) 58 (d) 29

Answer: (c) 58 — CH_3CH_2CHO.

103. The molecular ion of butanal appears at m/e:

(a) 72 (b) 58
(c) 86 (d) 43

Answer: (a) 72 — $CH_3CH_2CH_2CHO$.

104. Simple alkanes fragment by breaking:

(a) O–H bonds (b) C–H bonds only
(c) the nucleus (d) C–C bonds

Answer: (d) C–C bonds — Stated in the book.

105. The CH_3^+ fragment appears at m/e:

(a) 43 (b) 29
(c) 15 (d) 57

Answer: (c) 15 — 12 + 3.

106. The $C_2H_5^+$ fragment appears at m/e:

(a) 15 (b) 29
(c) 43 (d) 71

Answer: (b) 29 — 24 + 5.

107. The $C_3H_7^+$ fragment appears at m/e:

(a) 85 (b) 29
(c) 57 (d) 43

Answer: (d) 43 — 36 + 7.

108. The $C_4H_9^+$ fragment appears at m/e:

(a) 71 (b) 43
(c) 57 (d) 15

Answer: (c) 57 — 48 + 9.

109. The molecular ion of hexane appears at m/e:

(a) 86 (b) 85
(c) 71 (d) 100

Answer: (a) 86 — C_6H_{14}.

110. A peak at m/e 18 may be due to the loss of:

(a) carbon dioxide (b) water
(c) an ethyl group (d) a methyl group

Answer: (b) water — H_2O = 18.

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111. A peak at m/e 44 may be due to:

(a) ethyl (b) water
(c) methyl (d) carbon dioxide

Answer: (d) carbon dioxide — CO_2 = 44.

112. A compound with one chlorine atom gives M : [M+2] as:

(a) 1 : 1 (b) 3 : 1
(c) 9 : 6 (d) 1 : 2

Answer: (b) 3 : 1 — Cl-35 is three times as abundant.

113. A compound with two chlorine atoms gives M : [M+2] : [M+4] as:

(a) 3 : 1 : 1 (b) 1 : 2 : 1
(c) 9 : 6 : 1 (d) 1 : 1 : 1

Answer: (c) 9 : 6 : 1 — Stated in the book.

114. A compound with one bromine atom gives M : [M+2] as:

(a) 1 : 1 (b) 3 : 1
(c) 9 : 1 (d) 2 : 1

Answer: (a) 1 : 1 — The two isotopes are equally abundant.

115. A compound with two bromine atoms gives M : [M+2] : [M+4] as:

(a) 1 : 1 : 1 (b) 9 : 6 : 1
(c) 3 : 1 : 3 (d) 1 : 2 : 1

Answer: (d) 1 : 2 : 1 — Stated in the book.

116. The isotopes of bromine are:

(a) Br-35 and Br-37 (b) Br-79 and Br-81
(c) Br-80 and Br-82 (d) Br-78 and Br-79

Answer: (b) Br-79 and Br-81 — Stated in the book.

117. The mass spectrum of CH_3Br shows molecular ion peaks at:

(a) 79 and 81 (b) 50 and 52
(c) 94 and 96 (d) 15 and 29

Answer: (c) 94 and 96 — 15 + 79 and 15 + 81.

118. The mass spectrum of CH_3Cl shows molecular ion peaks at:

(a) 50 and 52 (b) 94 and 96
(c) 35 and 37 (d) 45 and 47

Answer: (a) 50 and 52 — 15 + 35 and 15 + 37.

119. Different compounds with the same molecular mass are told apart by:

(a) their molecular ion peak (b) their fragmentation patterns
(c) their [M+1] ratio only (d) their colour

Answer: (b) their fragmentation patterns — Stated in the book.

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